

UNIT-1: Electrostatic Fields

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Chapter 1

Electrostatic Fields

Electrostatics is a fascinating subject that has grown up in diverse areas of application. Electric power transmission, X-ray machines, and lightning protection are associated with strong electric fields and will require a knowledge of electrostatics to understand and design suitable equipment. The devices used in solid-state electronics are based on electrostatics. These include resistors, capacitors, and active devices such as bipolar and field effect transistors, which are based on control of electron motion by electrostatic fields. Almost all computer peripheral devices, with the exception of magnetic memory, are based on electrostatic fields. Touch pads, capacitance keyboards, cathode-ray tubes, liquid crystal displays, and electrostatic printers are typical examples. In medical work, diagnosis is often carried out with the aid of electrostatics, as incorporated in electrocardiograms, electroencephalograms, and other recordings of the electrical activity of organs including eyes, ears, and the stomach. In industry, electrostatics is applied in a variety of forms such as paint spraying, electrodeposition, electrochemical machining, and separation of fine particles. Electrostatics is used in agriculture to sort seeds, for direct spraying of plants, to measure the moisture content of crops, to spin cotton, and for speed-baking bread and smoking meat.

We begin our study of electrostatics by investigating the two fundamental laws governing electrostatic fields: (1) Coulomb's law and (2) Gauss's law. Both of these laws are based on experimental studies, and they are interdependent. Although Coulomb's law is applicable in finding the electric field due to any charge configuration, it is easier to use Gauss's law when charge distribution is symmetrical. Based on Coulomb's law, the concept of electric field intensity will be introduced and applied to cases involving point, line, surface, and volume charges. Special problems that can be solved with much effort using Coulomb's law will be solved with ease by applying Gauss's law. Throughout our discussion in this chapter, we will assume that the electric field is in a vacuum or free space.

COULOMB'S LAW AND FIELD INTENSITY

Coulomb's law is an experimental law formulated in 1785 by Charles Augustin de Coulomb, then a colonel in the French army. It deals with the force a point charge exerts on another point charge. By a *point charge* we mean a charge that is located on a body whose dimensions are much smaller than other relevant dimensions. For example, a collection of electric charges on a pinhead may be regarded as a point charge. Electrons are regarded as point charges. The polarity of charges may be positive or negative; like charges repel, while unlike charges attract. Charges are generally measured in coulombs (C). One coulomb is approximately equivalent to 6×10^{18} electrons; it is a very large unit of charge because one electron charge $e = -1.6019 \times 10^{-19}$ C.

Coulomb's law states that the force F between two point charges Q_1 and Q_2 is:

1. Along the line joining them
2. Directly proportional to the product $Q_1 Q_2$ of the charges
3. Inversely proportional to the square of the distance R between them.

Expressed mathematically,

$$F = \frac{kQ_1 Q_2}{R^2} \quad (1.1)$$

where k is the proportionality constant whose value depends on the choice of system of units. In SI units, charges Q_1 and Q_2 are in coulombs (C), the distance R is in meters (m), and the force F is in newtons (N) so that $k = 1/4\pi\epsilon_0$. The constant ϵ_0 is known as the *permittivity of free space* (in farads per meter) and has the value

$$\epsilon_0 = 8.854 \times 10^{-12} \approx \frac{10^{-9}}{36\pi} \text{ F/m} \quad (1.2)$$

or

$$k = \frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \text{ m/F} \quad (1.3)$$

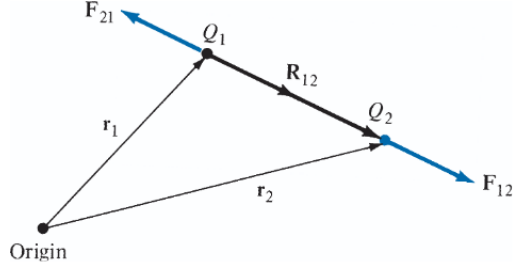


Figure 1.1: Coulomb vector force on point charges Q_1 and Q_2 .

Thus eq. (1.1) becomes

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \quad (1.4)$$

If point charges Q_1 and Q_2 are located at points having position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, then the force $\mathbf{F}_{12}$ on Q_2 due to Q_1 , shown in Figure 1.1, is given by

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \mathbf{a}_{R_{12}} \quad (1.5)$$

where

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1 \quad (1.6)$$

$$R = |\mathbf{R}_{12}| \quad (1.7)$$

$$\mathbf{a}_{R_{12}} = \frac{\mathbf{R}_{12}}{R} \quad (1.8)$$

By substituting eq. (1.8) into eq. (1.5), we may write eq. (1.5) as

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^3} \mathbf{R}_{12} \quad (1.9)$$

or

$$\mathbf{F}_{12} = \frac{Q_1 Q_2 (\mathbf{r}_2 - \mathbf{r}_1)}{4\pi\epsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|^3} \quad (1.10)$$

It is worthwhile to note that

1. As shown in Figure 1.1, the force $\mathbf{F}_{21}$ on Q_1 due to Q_2 is given by

$$\mathbf{F}_{21} = |\mathbf{F}_{12}| \mathbf{a}_{R_{21}} = |\mathbf{F}_{12}| (-\mathbf{a}_{R_{12}})$$

or

$$\mathbf{F}_{21} = -\mathbf{F}_{12} \quad (1.11)$$

since

$$\mathbf{a}_{R_{21}} = -\mathbf{a}_{R_{12}}$$

2. Like charges (charges of the same sign) repel each other, while unlike charges attract. This is illustrated in Figure 1.2.

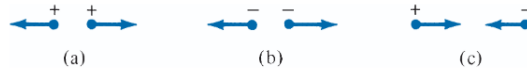


Figure 1.2: (a), (b) Like charges repel. (c) Unlike charges attract

3. The distance R between the charged bodies Q_1 and Q_2 must be large compared with the linear dimensions of the bodies; that is, Q_1 and Q_2 must be point charges.
4. Q_1 and Q_2 must be static (at rest).
5. The signs of Q_1 and Q_2 must be taken into account. For like charges, $Q_1 Q_2 > 0$. For unlike charges, $Q_1 Q_2 < 0$.
6. Charges cannot be created or destroyed; the quantity of total charge remains constant.

If we have more than two point charges, we can use the *principle of superposition* to determine the force on a particular charge. The principle states that if there are N charges $Q_1, Q_2, \dots, Q_N$ located, respectively, at points with position vectors $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$, the resultant force $\mathbf{F}$ on a charge Q located at point $\mathbf{r}$ is the vector sum of the forces exerted on Q by each of the charges $Q_1, Q_2, \dots, Q_N$. Hence,

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots + \mathbf{F}_N \quad (1.12)$$

$$\mathbf{F} = \frac{QQ_1(\mathbf{r} - \mathbf{r}_1)}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_1|^3} + \frac{QQ_2(\mathbf{r} - \mathbf{r}_2)}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{QQ_N(\mathbf{r} - \mathbf{r}_N)}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_N|^3} \quad (1.13)$$

or

$$\mathbf{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \quad (1.14)$$

We can now introduce the concept of *electric field intensity*.

The electric field intensity (or electric field strength) $\mathbf{E}$ is the force that a unit positive charge experiences when placed in an electric field.

Thus,

$$\mathbf{E} = \lim_{Q \rightarrow 0} \frac{\mathbf{F}}{Q} \quad (1.15)$$

or simply

$$\mathbf{E} = \frac{\mathbf{F}}{Q} \quad (1.16)$$

For $Q > 0$, the electric field intensity $\mathbf{E}$ is obviously in the direction of the force $\mathbf{F}$ and is measured in newtons per coulomb or volts per meter. The electric field intensity at point $\mathbf{r}$ due to a point charge located at $\mathbf{r}'$ is readily obtained from eqs (1.10) and (1.16) as

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3} \quad (1.17)$$

or simply

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \quad (1.18)$$

For N point charges $Q_1, Q_2, \dots, Q_N$ located at $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$, the electric field intensity at point $\mathbf{r}$ is obtained from eqs. (1.14) and (1.16) as

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots + \mathbf{E}_N \quad (1.19)$$

$$\mathbf{E} = \frac{Q_1(\mathbf{r} - \mathbf{r}_1)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|^3} + \frac{Q_2(\mathbf{r} - \mathbf{r}_2)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{Q_N(\mathbf{r} - \mathbf{r}_N)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_N|^3} \quad (1.20)$$

or

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \quad (1.21)$$

1.1 Electric Fields Due to Continuous Charge Distributions

So far we have considered only forces and electric fields due to point charges, which are essentially charges occupying very small physical space. It is also possible to have continuous charge distribution along a line, on a surface, or in a volume, as illustrated in Fig.1.3.

It is customary to denote the line charge density, surface charge density, and volume charge density by ρ_L (in C/m), ρ_S (in C/m²), and ρ_v (in C/m³), respectively.

These must not be confused with ρ (without subscript), used for radial distance in cylindrical coordinates.

The charge element dQ and the total charge Q due to these charge distributions are obtained from Fig 1.3 as

$$dQ = \rho_L dl \quad \Rightarrow \quad Q = \int_L \rho_L dl \quad (1.22)$$

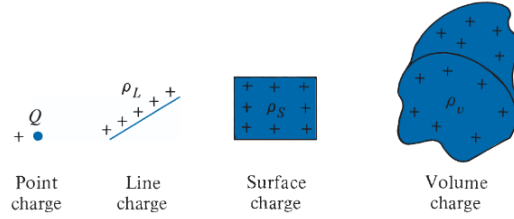


Figure 1.3: Various charge distributions and charge elements

$$dQ = \rho_S dS \quad \Rightarrow \quad Q = \int_S \rho_S dS \quad (1.23)$$

$$dQ = \rho_v dV \quad \Rightarrow \quad Q = \int_V \rho_v dV \quad (1.24)$$

The electric field intensity due to each of the charge distributions ρ_L , ρ_S , and ρ_v may be regarded as the summation of the field contributed by the numerous point charges making up the charge distribution. We treat dQ as a point charge. Thus by replacing Q in eq. (1.17) with the charge element $dQ = \rho_L dl$, $\rho_S dS$, or $\rho_v dV$ and integrating, we get

$$\mathbf{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (1.25)$$

$$\mathbf{E} = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (1.26)$$

$$\mathbf{E} = \int_V \frac{\rho_v dV}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (1.27)$$

It should be noted that R^2 and $\mathbf{a}_R$ vary as the integrals in eqs.(1.23) to (1.27) are evaluated. We shall now apply these formulas to some specific charge distributions.

1.2 Electric Flux Density

The flux due to the electric field $\mathbf{E}$ can be calculated by using the general definition of flux. For practical reasons, however, this quantity is not usually considered to be the most useful flux in electrostatics. Also, eqs. (1.17) to (1.27) show that the electric field intensity is dependent on the medium in which the charge is placed. Suppose a new vector field $\mathbf{D}$ is defined by

$$\mathbf{D} = \varepsilon_0 \mathbf{E} \quad (1.28)$$

We use the definition of electric flux Ψ in terms of $\mathbf{D}$, namely,

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S} \quad (1.29)$$

In SI units, one line of electric flux emanates from +1 C and terminates on -1 C. Therefore, the electric flux is measured in coulombs. Hence, the vector field $\mathbf{D}$ is called the *electric flux density* and is measured in coulombs per square meter. For historical reasons, the electric flux density is also called *electric displacement*.

From eq. (1.28), it is apparent that all the formulas derived for $\mathbf{E}$ from Coulomb's law can be used in calculating $\mathbf{D}$, except that we have to multiply those formulas by ε_0 . For example, for an infinite sheet of charge gives

$$\mathbf{D} = \frac{\rho_s}{2} \mathbf{a}_n \quad (1.30)$$

and for a volume charge distribution, eqs. (1.27) and (1.28) give

$$\mathbf{D} = \int_v \frac{\rho_v dv}{4\pi R^2} \mathbf{a}_R \quad (1.31)$$

Note from eqs. (1.30) and (1.31) that $\mathbf{D}$ is a function of charge and position only; it is independent of the medium.

1.3 Gauss's Law—Maxwell's Equation

Gauss's law constitutes one of the fundamental laws of electromagnetism.

Gauss's law states that the total electric flux Ψ through any closed surface is equal to the total charge enclosed by that surface.

Thus,

$$\Psi = Q_{\text{enc}} \quad (1.32)$$

that is,

$$\Psi = \oint_S d\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_v \rho_v dv \quad (1.33)$$

or

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv \quad (1.34)$$

By applying the divergence theorem to the middle term in eq. (1.34), we have

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{D} dv \quad (1.35)$$

Comparing the two volume integrals in eqs. (1.34) and (1.35) results in

$$\rho_v = \nabla \cdot \mathbf{D} \quad (1.36)$$

which is the first of the four Maxwell's equations to be derived. Equation (1.36) states that the volume charge density is the same as the divergence of the electric flux density. It is equivalent to Coulomb's law of force between point charges.

Note that:

1. Equations (1.34) and (1.36) are basically stating Gauss's law in different ways: eq. (1.34) is the integral form, whereas eq. (1.36) is the differential or point form of Gauss's law. Equation (1.36) is sometimes called the source equation.
2. Gauss's law is an alternative statement of Coulomb's law; proper application of the divergence theorem to Coulomb's law results in Gauss's law.
3. Gauss's law provides an easy means of finding $\mathbf{E}$ or $\mathbf{D}$ for symmetrical charge distributions such as a point charge, an infinite line charge, an infinite cylindrical surface charge, and a spherical distribution of charge. A continuous charge distribution has rectangular symmetry if it depends only on x (or y or z), cylindrical symmetry if it depends only on ρ , or spherical symmetry if it depends only on r (independent of θ and ϕ). It must be stressed that whether the charge distribution is symmetric or not, Gauss's law always holds.

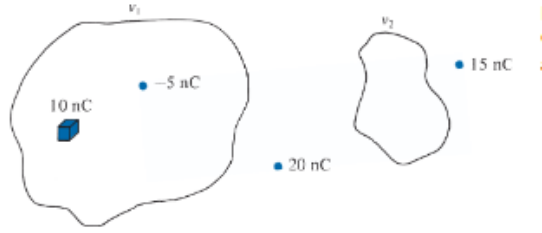


Figure 1.4: Illustration of Gauss's law: flux leaving v_1 is 5 nC and that leaving v_2 is 0 C.

Consider the charge distribution in Fig.1.4 where v_1 and v_2 are closed surfaces (or volumes). The total flux leaving v_1 is $10 - 5 = 5$ nC because only 10 nC and -5 nC charges are enclosed by v_1 . Although charges 20 nC and 15 nC outside v_1 do contribute to the flux crossing v_1 , the net flux crossing v_1 , according to Gauss's law, is irrespective of those charges outside v_1 . Similarly, the total flux leaving v_2 is zero because no charge is enclosed by v_2 . Thus we see that Gauss's law, $\Psi = Q_{\text{enc}}$, is still obeyed even though the charge distribution is not symmetric. However, we cannot use the law to determine $\mathbf{E}$ or $\mathbf{D}$ when the charge distribution is not symmetric; we must resort to Coulomb's law to determine $\mathbf{E}$ or $\mathbf{D}$ in that case.

1.4 Applications of Gauss's Law

The procedure for applying Gauss's law to calculate the electric field involves first knowing whether symmetry exists. Once it has been found that symmetric charge distribution exists, we construct a mathematical closed surface (known as a *Gaussian surface*). The surface is chosen such that $\mathbf{D}$ is normal or tangential to the Gaussian surface. When $\mathbf{D}$ is normal to the surface, $\mathbf{D} \cdot d\mathbf{S} = D dS$ because $\mathbf{D}$ is perpendicular to the surface. When $\mathbf{D}$ is tangential to the surface, $\mathbf{D} \cdot d\mathbf{S} = 0$. Thus we must choose a surface that has some of the symmetry exhibited by the charge distribution. The choice of an appropriate Gaussian surface, where there is symmetry in the charge distribution, comes from intuitive reasoning and a slight degree of maturity in the application of Coulomb's law. We shall now apply these basic ideas to the following cases.

A. Point Charge

Suppose a point charge Q is located at the origin. To determine $\mathbf{D}$ at a point P , it is easy to see that choosing a spherical surface containing P will satisfy symmetry conditions. Thus, a spherical surface centered at the origin is the Gaussian surface in this case (Fig 1.5).

Since $\mathbf{D}$ is everywhere normal to the Gaussian surface, that is, $\mathbf{D} = D_r \mathbf{a}_r$, applying Gauss's law ($\Psi = Q_{\text{enc}}$) gives

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = D_r \oint_S dS = D_r 4\pi r^2 \quad (1.37)$$

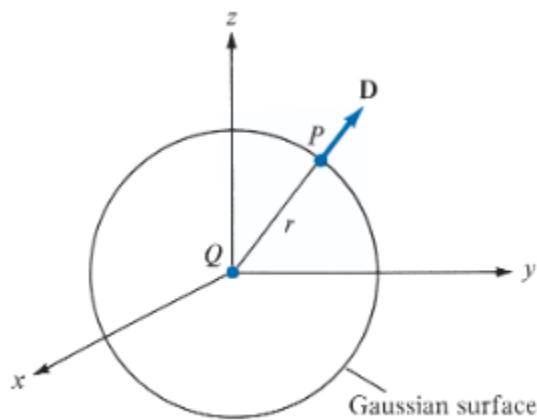


Figure 1.5: Gaussian surface about a point charge

B. Infinite Line Charge

Suppose the infinite line of uniform charge ρ_L C/m lies along the z -axis. To determine $\mathbf{D}$ at a point P , we choose a cylindrical surface containing P to satisfy the symmetry condition as shown in Fig 1.6. The electric flux density $\mathbf{D}$ is constant on and normal to the cylindrical Gaussian surface; that is,

$$\mathbf{D} = D_\rho \mathbf{a}_\rho.$$

If we apply Gauss's law to an arbitrary length ℓ of the line,

$$\rho_L \ell = Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = D_\rho \oint_S dS = D_\rho 2\pi\rho\ell \quad (1.38)$$

where

$$\oint_S dS = 2\pi\rho\ell$$

is the surface area of the cylindrical Gaussian surface.

Note that $\oint_S \mathbf{D} \cdot d\mathbf{S}$ evaluated on the top and bottom surfaces of the cylinder is zero, since $\mathbf{D}$ has no z -component; that means that $\mathbf{D}$ is tangential to those surfaces.

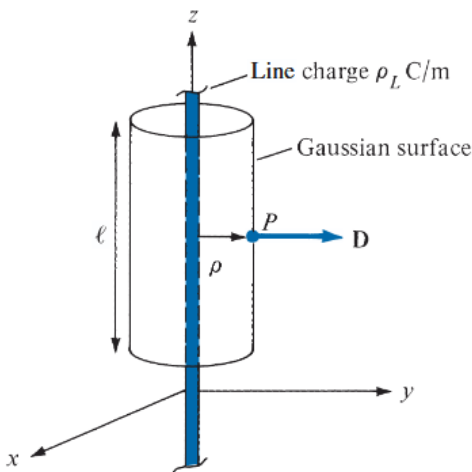


Figure 1.6: Gaussian surface about an infinite line

C. Infinite Sheet of Charge

Consider an infinite sheet of uniform charge ρ_S C/m² lying on the $z = 0$ plane. To determine $\mathbf{D}$ at point P , we choose a rectangular box that is cut symmetrically by the sheet of charge and has two of its faces parallel to the (Fig 1.7). As $\mathbf{D}$ is normal to the sheet, $\mathbf{D} = D_z \mathbf{a}_z$, and applying Gauss's law gives

$$\rho_S \int_S dS = Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = D_z \left[\int_{\text{top}} dS + \int_{\text{bottom}} dS \right] \quad (1.39)$$

Note that $\mathbf{D} \cdot d\mathbf{S}$ evaluated on the sides of the box is zero because $\mathbf{D}$ has no components along $\mathbf{a}_x$ and $\mathbf{a}_y$. If the top and bottom areas of the box each have area A , eq. (1.39) becomes

$$\rho_S A = D_z (A + A) \quad (1.40)$$

and thus

$$\mathbf{D} = \frac{\rho_S}{2} \mathbf{a}_z \quad (1.41)$$

or

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_0} = \frac{\rho_S}{2\varepsilon_0} \mathbf{a}_z \quad (1.42)$$

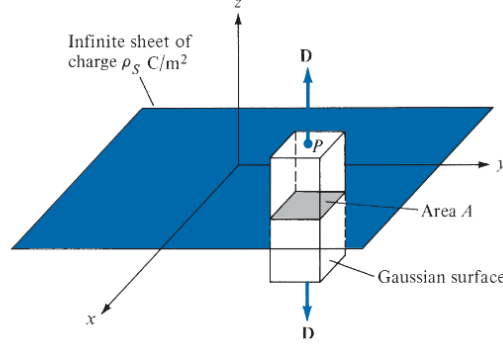


Figure 1.7: Gaussian surface about an infinite line sheet of charge

D. Uniformly Charged Sphere

Consider a sphere of radius a with a uniform charge density ρ_v C/m³. To determine $\mathbf{D}$ everywhere, we construct Gaussian surfaces for the cases $r \leq a$ and $r \geq a$ separately. Since the charge has spherical symmetry, a spherical surface is an appropriate Gaussian surface.

For $r \leq a$, (Fig 1.8a) the total charge enclosed by the spherical surface of radius r is

$$Q_{\text{enc}} = \int_v \rho_v dv = \rho_v \int_0^{2\pi} \int_0^\pi \int_0^r r'^2 \sin \theta dr' d\theta d\phi = \rho_v \frac{4\pi r^3}{3} \quad (1.43)$$

and

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = D_r \oint_S dS = D_r 4\pi r^2 \quad (1.44)$$

Hence, $\Psi = Q_{\text{enc}}$ gives

$$D_r 4\pi r^2 = \frac{4\pi r^3}{3} \rho_v \quad (1.45)$$

or

$$\mathbf{D} = \frac{r}{3} \rho_v \mathbf{a}_r, \quad 0 < r \leq a \quad (1.46)$$

For $r \geq a$, (Fig 1.8b) the charge enclosed is the entire charge:

$$Q_{\text{enc}} = \int_v \rho_v dv = \rho_v \frac{4\pi a^3}{3} \quad (1.47)$$

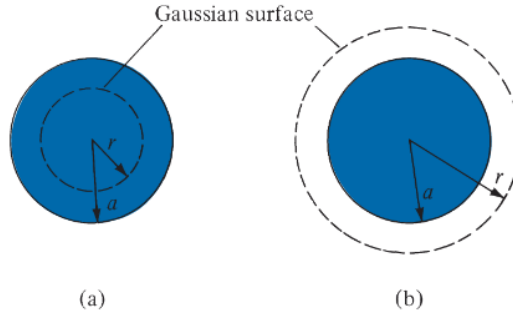


Figure 1.8: Gaussian surface for a uniformly charged sphere when (a) $r \leq a$ and (b) $r \geq a$

while

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = D_r 4\pi r^2 \quad (1.48)$$

just as in eq. (1.44). Hence,

$$D_r 4\pi r^2 = \frac{4}{3}\pi a^3 \rho_v \quad (1.49)$$

or

$$\mathbf{D} = \frac{a^3}{3r^2} \rho_v \mathbf{a}_r, \quad r \geq a \quad (1.50)$$

Thus from eqs. (1.46) and (1.50), $\mathbf{D}$ everywhere is given by

$$\mathbf{D} = \begin{cases} \frac{r}{3} \rho_v \mathbf{a}_r, & 0 < r \leq a \\ \frac{a^3}{3r^2} \rho_v \mathbf{a}_r, & r \geq a \end{cases} \quad (1.51)$$

and $|\mathbf{D}|$ is as sketched in Fig 1.9.

Notice from eqs. (1.37), (1.38), (1.39), and (1.44) that the ability to take $\mathbf{D}$ out of the integral sign is the key to finding $\mathbf{D}$ using Gauss's law. In other words, $\mathbf{D}$ must be constant on the Gaussian surface.

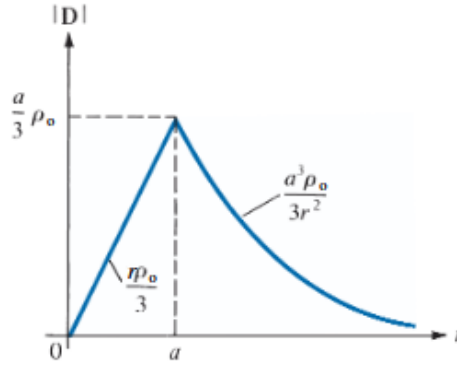


Figure 1.9: Sketch of D against r for a uniformly charged sphere

1.5 Electric Potential

From our discussions in the preceding sections, we can obtain the electric field intensity $\mathbf{E}$ due to a charge distribution from Coulomb's law in general or, when the charge distribution is symmetric, from Gauss's law. Another way of obtaining $\mathbf{E}$ is from the electric scalar potential V , to be defined in this section. In a sense, this way of finding $\mathbf{E}$ is easier because it is easier to handle scalars than vectors.

Suppose we wish to move a point charge Q from point A to point B in an electric field $\mathbf{E}$ as shown in Fig.1.10. From Coulomb's law, the force on Q is $\mathbf{F} = Q\mathbf{E}$ so that the work done in displacing the charge by $d\mathbf{l}$ is

$$dW = -\mathbf{F} \cdot d\mathbf{l} = -Q\mathbf{E} \cdot d\mathbf{l} \quad (1.52)$$

The negative sign indicates that the work is being done by an external agent. Thus the total work done, or the potential energy required, in moving Q from A to B is

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} \quad (1.53)$$

Dividing W by Q in eq. (1.53) gives the potential energy per unit charge. This quantity, denoted by V_{AB} , is known as the potential difference between points A and B . Thus

$$V_{AB} = \frac{W}{Q} = - \int_A^B \mathbf{E} \cdot d\mathbf{l} \quad (1.54)$$

Note that:

1. In determining V_{AB} , A is the initial point while B is the final point.
2. If V_{AB} is negative, there is a loss in potential energy in moving Q from A to B ; this implies that the work is being done by the field. However, if V_{AB} is positive, there is a gain in potential energy in the movement; an external agent performs the work.
3. V_{AB} is independent of the path taken (to be shown later).
4. V_{AB} is measured in joules per coulomb, commonly referred to as volts (V).

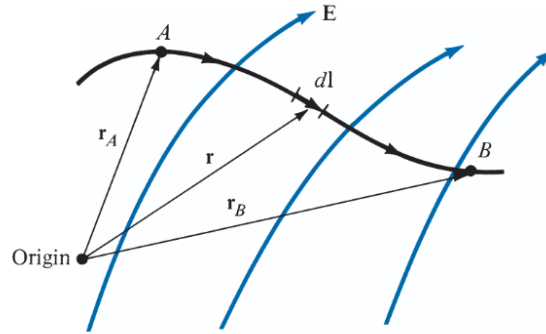


Figure 1.10: Displacement of point charge Q in an electrostatic field E .

As an example, if the electric field in Fig.1.10 is due to a point charge Q located at the origin, then

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \quad (1.55)$$

so eq. (1.54) becomes

$$V_{AB} = - \int_{r_A}^{r_B} \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \cdot d\mathbf{r} \mathbf{a}_r \quad (1.56)$$

$$= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \quad (1.57)$$

or

$$V_{AB} = V_B - V_A \quad (1.58)$$

where V_B and V_A are the potentials (or absolute potentials) at B and A , respectively. Thus the potential difference V_{AB} may be regarded as the potential at B with reference to A .

In problems involving point charges, it is customary to choose infinity as reference; that is, we assume the potential at infinity is zero. Thus if $V_A = 0$ as $r_A \rightarrow \infty$, in eq.(1.58), the potential at any point ($r_B \rightarrow r$) due to a point charge Q located at the origin is

$$V = \frac{Q}{4\pi\epsilon_0 r} \quad (1.59)$$

Note from eq. (1.56) that because $\mathbf{E}$ points in the radial direction, any contribution from a displacement in the θ or ϕ direction is eliminated by the dot product $\mathbf{E} \cdot d\mathbf{l} = E \cos \alpha dl = E dr$, where α is the angle between $\mathbf{E}$ and $d\mathbf{l}$. Hence the potential difference V_{AB} is independent of the path as asserted earlier.

In general, vectors whose line integral does not depend on the path of integration are called *conservative*. Thus, $\mathbf{E}$ is conservative.

The potential at any point is the potential difference between that point and a chosen point (or reference point) at which the potential is zero. In other words, if one assumes zero potential at infinity, the potential at a distance r from the point charge is the work done per unit charge by an external agent in transferring a test charge from infinity to that point. Thus

$$V = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} \quad (1.60)$$

If the point charge Q eq.(1.59) is not located at the origin but at a point whose position vector is $\mathbf{r}'$, the potential $V(x, y, z)$ or simply $V(\mathbf{r})$ at $\mathbf{r}$ becomes

$$V(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} \quad (1.61)$$

We have considered the electric potential due to a point charge. The same basic ideas apply to other types of charge distribution because any charge distribution can be regarded as consisting of point charges. The superposition principle applies to potentials also.

For point charges $Q_1, Q_2, \dots, Q_n$ located at positions $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$, the potential at $\mathbf{r}$ is

$$V(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|} + \frac{Q_2}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_2|} + \dots + \frac{Q_n}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_n|} \quad (1.62)$$

or

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{Q_k}{|\mathbf{r} - \mathbf{r}_k|} \quad (1.63)$$

For continuous charge distributions, we replace Q_k in eq. (1.63) with the charge element $\rho_L dl$, $\rho_S dS$, or $\rho_v dv$ and the summation becomes an integration. Thus the potential at $\mathbf{r}$ becomes

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_L \frac{\rho_L(\mathbf{r}') dl'}{|\mathbf{r} - \mathbf{r}'|} \quad (1.64)$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\rho_S(\mathbf{r}') dS'}{|\mathbf{r} - \mathbf{r}'|} \quad (1.65)$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_v \frac{\rho_v(\mathbf{r}') dv'}{|\mathbf{r} - \mathbf{r}'|} \quad (1.66)$$

where the primed coordinates are used to denote the source point location and the unprimed coordinates refer to the field point.

1. We recall that in obtaining eq.(1.59) to (1.66), the zero potential (reference point) has been chosen arbitrarily at infinity. If any other point is chosen as reference, eq.(1.61) becomes

$$V = \frac{Q}{4\pi\epsilon_0 r} + C \quad (1.67)$$

where C is a constant determined at the chosen point of reference. The same idea applies to eqs.(1.59) to (1.66).

2. The potential at a point can be determined in two ways depending on whether the charge distribution or $\mathbf{E}$ is known. If the charge distribution is known, we use one of eqs. (1.61) to (1.67). If $\mathbf{E}$ is known, we simply use

$$V = - \int \mathbf{E} \cdot d\mathbf{l} + C \quad (1.68)$$

The potential difference V_{AB} can be found generally from

$$V_{AB} = V_B - V_A = - \int_A^B \mathbf{E} \cdot d\mathbf{l} = \frac{W}{Q} \quad (1.69)$$

1.6 Relationship Between $\mathbf{E}$ and V —Maxwell's Equation

As shown in the preceding section, the potential difference between points A and B is independent of the path taken. Hence,

$$V_{BA} = -V_{AB}$$

that is,

$$V_{BA} + V_{AB} = \oint_L \mathbf{E} \cdot d\mathbf{l} = 0$$

or

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0 \quad (1.70)$$

This shows that the line integral of $\mathbf{E}$ along a closed path must be zero (Fig 1.11). Physically, this implies that no net work is done in moving a charge along a closed path in an electrostatic field.

Applying Stokes's theorem to eq. (1.70) gives

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = 0 \quad (1.71)$$

or

$$\nabla \times \mathbf{E} = 0 \quad (1.72)$$

Any vector field that satisfies eq. (1.70) or (1.72) is said to be *conservative*, or *irrotational*. In other words, vectors whose line integral does not depend on the path of integration are called conservative vector fields. Thus an electrostatic field is a conservative field. Equation (1.70) or (1.72) is referred to as Maxwell's equation for static electric fields. Equation (1.70) is the integral form, and eq. (1.72) is the differential form; they both depict the conservative nature of an electrostatic field.

From the way we defined potential, $V = -\int \mathbf{E} \cdot d\mathbf{l}$, it follows that

$$dV = -\mathbf{E} \cdot d\mathbf{l} = -E_x dx - E_y dy - E_z dz \quad (1.73)$$

But from calculus of multivariable functions, a total change in $V(x, y, z)$ is the sum of partial changes with respect to x , y , and z :

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \quad (1.74)$$

Comparing the two expressions for dV , we obtain

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z} \quad (1.75)$$

Thus,

$$\mathbf{E} = -\nabla V \quad (1.76)$$

That is, the electric field intensity is the gradient of V . The negative sign shows that the direction of $\mathbf{E}$ is opposite to the direction in which V increases; $\mathbf{E}$ is directed from higher to lower levels of V .

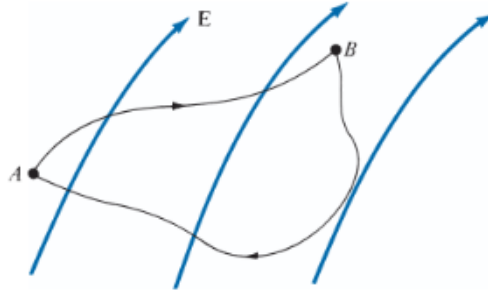


Figure 1.11: The conservative nature of an electrostatic field

Since the curl of the gradient of a scalar function is always zero ($\nabla \times \nabla V = 0$), eq. (1.72) obviously implies that $\mathbf{E}$ must be a gradient of some scalar function. Thus eq. (1.76) could have been obtained from eq. (1.72).

Equation (1.76) shows another way to obtain the electric field apart from using Coulomb's or Gauss's law. That is, if the potential field V is known, $\mathbf{E}$ can be found using eq. (1.76).

One may wonder how one function V can possibly contain all the information that the three components of $\mathbf{E}$ carry. The three components of $\mathbf{E}$ are not independent of one another; they are explicitly interrelated by the condition $\nabla \times \mathbf{E} = 0$. The potential formulation exploits this feature to maximum advantage, reducing a vector problem to a scalar one.

1.7 Poisson's and Laplace's Equations

Poisson's and Laplace's equations are easily derived from Gauss's law (for a linear, isotropic material medium):

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\varepsilon \mathbf{E}) = \rho_v \quad (1.77)$$

and

$$\mathbf{E} = -\nabla V \quad (1.78)$$

Substituting eq. (1.78) into eq. (1.77) gives

$$\nabla \cdot (-\varepsilon \nabla V) = \rho_v \quad (1.79)$$

for an inhomogeneous medium. For a homogeneous medium, eq. (1.79) becomes

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad (1.80)$$

This is known as *Poisson's equation*. A special case of this equation occurs when $\rho_v = 0$ (i.e., for a charge-free region). Equation (1.80) then becomes

$$\nabla^2 V = 0 \quad (1.81)$$

which is known as *Laplace's equation*. Note that in taking ε out of the left-hand side of eq. (1.79) to obtain eq. (1.80), we have assumed that ε is constant throughout the region, which V is defined; for an inhomogeneous region, ε is not constant and eq. (1.80) does not follow from eq. (1.79).

Recall that the Laplacian operator $\nabla^2 V$ in Cartesian, cylindrical, or spherical coordinates, respectively, is given by

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad (1.82)$$

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad (1.83)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0 \quad (1.84)$$

depending on the coordinate variables used to express V , that is, $V(x, y, z)$, $V(\rho, \phi, z)$, or $V(r, \theta, \phi)$.

Poisson's equation in those coordinate systems may be obtained by simply replacing zero on the right-hand side of eqs. (1.82)–(1.84) with $-\rho_v/\varepsilon$.

Laplace's equation is of primary importance in solving electrostatic problems involving a set of conductors maintained at different potentials. Examples include capacitors and vacuum tube diodes. Laplace's and Poisson's equations are not only

useful in electrostatics; they are used in various other field problems. For example, V would represent magnetic potential in magnetostatics, temperature in heat conduction, stress function in fluid flow, and pressure head in seepage.

1.8 General Procedures for Solving Poisson's or Laplace's Equation

The following general procedure may be taken in solving a given boundary-value problem involving Poisson's or Laplace's equation:

1. Solve Laplace's equation (if $\rho_v = 0$) or Poisson's equation (if $\rho_v \neq 0$) using either (a) direct integration when V is a function of one variable or (b) separation of variables if V is a function of more than one variable. The solution at this point is not unique but is expressed in terms of unknown integration constants to be determined.
2. Apply the boundary conditions to determine a unique solution for V . Imposing the given boundary conditions makes the solution unique.
3. Having obtained V , find $\mathbf{E}$ using $\mathbf{E} = -\nabla V$, $\mathbf{D}$ from $\mathbf{D} = \epsilon\mathbf{E}$, and $\mathbf{J}$ from $\mathbf{J} = \sigma\mathbf{E}$.
4. If required, find the charge Q induced on a conductor using

$$Q = \int_S \rho_S dS,$$

where $\rho_S = D_n$ and D_n is the component of $\mathbf{D}$ normal to the conductor.

If necessary, the capacitance of two conductors can be found using

$$C = \frac{Q}{V},$$

or the resistance of an object can be found using

$$R = \frac{V}{I}, \quad \text{where } I = \int_S \mathbf{J} \cdot d\mathbf{S}.$$

Solving Laplace's (or Poisson's) equation, as in step 1, is not always as complicated as it may seem. In some cases, the solution may be obtained by mere inspection of the problem. Also, a solution may be checked by going backward and finding out if it satisfies both Laplace's (or Poisson's) equation and the prescribed boundary condition.

1.8.1 CAPACITANCE

Generally speaking, to have a capacitor we must have two (or more) conductors carrying equal but opposite charges. This implies that all the flux lines leaving one conductor must necessarily terminate at the surface of the other conductor. The conductors are sometimes referred to as the *plates* of the capacitor. The plates may be separated by free space or a dielectric.

Consider the two-conductor capacitor of Figure 1.12. The conductors are maintained at a potential difference V given by

$$V = V_1 - V_2 = - \int_2^1 \mathbf{E} \cdot d\mathbf{l} \quad (1.85)$$

where $\mathbf{E}$ is the electric field existing between the conductors and conductor 1 is assumed to carry a positive charge. (Note that the $\mathbf{E}$ field is always normal to the conducting surfaces.)

We define the *capacitance* C of the capacitor as the ratio of the magnitude of the charge on one of the plates to the potential difference between them; that is,

$$C = \frac{Q}{V} = \frac{\varepsilon \int_S \mathbf{E} \cdot d\mathbf{S}}{\int_L \mathbf{E} \cdot d\mathbf{l}} \quad (1.86)$$

The negative sign before $V = - \int_L \mathbf{E} \cdot d\mathbf{l}$ has been dropped because we are interested in the absolute value of V . The capacitance C is a physical property of the capacitor and is measured in farads (F). Most capacitances are practically much smaller than a farad and are specified in microfarads (μF) or picofarads (pF). We can use eq. 1.86 to obtain C for any given two-conductor capacitance by following either of these methods:

1. Assuming Q and determining V in terms of Q (involving Gauss's law)

$$C = \frac{Q \text{ (assume)}}{V \text{ (find)}}$$

2. Assuming V and determining Q in terms of V (involving solving Laplace's equation)

$$C = \frac{Q \text{ (find)}}{V \text{ (assume)}}$$

1. Choose a suitable coordinate system.
2. Let the two conducting plates carry charges $+Q$ and $-Q$.

3. Determine $\mathbf{E}$ by using Coulomb's or Gauss's law and find V from

$$V = - \int_L \mathbf{E} \cdot d\mathbf{l}.$$

The negative sign may be ignored in this case because we are interested in the absolute value of V .

4. Finally, obtain C from $C = Q/V$.

We will now apply this mathematically attractive procedure to determine the capacitance of some important two-conductor configurations.

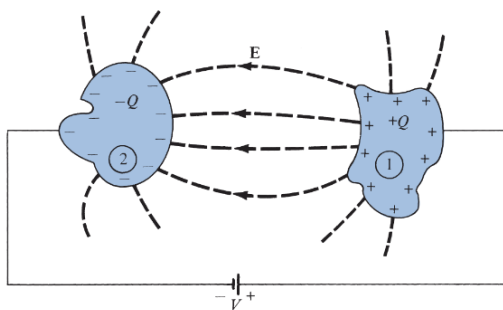
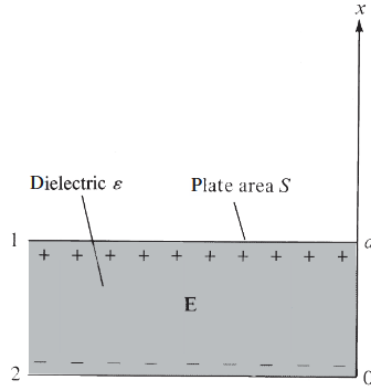


Figure 1.12: A two-conductor capacitor.

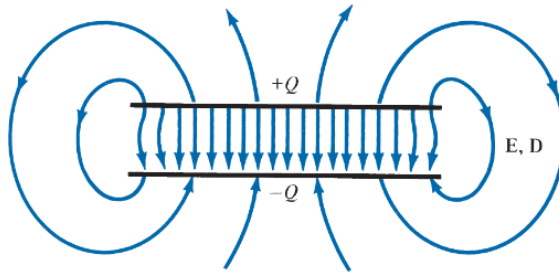
A. Parallel-Plate Capacitor

Consider the parallel-plate capacitor of Figure 1.13(a). Suppose that each of the plates has an area S and they are separated by a distance d . We assume that plates 1 and 2, respectively, carry charges $+Q$ and $-Q$ uniformly distributed on them so that

$$\rho_s = \frac{Q}{S} \tag{1.87}$$



(a)



(b)

Figure 1.13: (a) Parallel-plate capacitor. (b) Fringing effect due to a parallel-plate capacitor

An ideal parallel-plate capacitor is one in which the plate separation d is very small compared with the dimensions of the plates. Assuming such an ideal case, the fringing field at the edge of the plates, as illustrated in Figure 1.13(b), can be ignored so that the field between them is considered uniform. If the space between the plates is filled with a homogeneous dielectric with permittivity ε and we ignore flux fringing at the edges of the plates, from a known equation $\mathbf{D} = -\rho_s \mathbf{a}_x$,

$$\mathbf{E} = \frac{\rho_s}{\varepsilon}(-\mathbf{a}_x) = -\frac{Q}{\varepsilon S} \mathbf{a}_x \quad (1.88)$$

Hence,

$$V = -\int_2^1 \mathbf{E} \cdot d\mathbf{l} = -\int_0^d \left(-\frac{Q}{\varepsilon S} \mathbf{a}_x \right) \cdot \mathbf{a}_x dx = \frac{Qd}{\varepsilon S} \quad (1.89)$$

and thus for a parallel-plate capacitor

$$C = \frac{Q}{V} = \frac{\varepsilon S}{d} \quad (1.90)$$

This formula offers a means of measuring the dielectric constant ε_r of a given dielectric. By measuring the capacitance C of a parallel-plate capacitor with the space between the plates filled with the dielectric and the capacitance C_0 with air between the plates, we find ε_r from

$$\varepsilon_r = \frac{C}{C_0} \quad (1.91)$$

Using Energy density eq, it can be shown that the energy stored in a capacitor is given by

$$W_E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C} \quad (1.92)$$

To verify this for a parallel-plate capacitor, we substitute eq.1.88 into equation of energy density and obtain

$$W_E = \frac{1}{2} \int_V \varepsilon E^2 dv = \frac{1}{2} \varepsilon \left(\frac{Q^2}{\varepsilon^2 S^2} \right) Sd \quad (1.93)$$

$$= \frac{Q^2}{2} \left(\frac{d}{\varepsilon S} \right) = \frac{Q^2}{2C} = \frac{1}{2}QV \quad (1.94)$$

as expected.

B. Coaxial Capacitor

A coaxial capacitor is essentially a coaxial cable or coaxial cylindrical capacitor. Consider length L of two coaxial conductors of inner radius a and outer radius b ($b > a$) as shown in Figure 1.14. Let the space between the conductors be filled with a homogeneous dielectric with permittivity ε . We assume that conductors 1 and 2, respectively, carry $+Q$ and $-Q$ uniformly distributed on them. By applying Gauss's law to an arbitrary Gaussian cylindrical surface of radius ρ ($a < \rho < b$), we obtain

$$Q = \varepsilon \oint_S \mathbf{E} \cdot d\mathbf{S} = \varepsilon E_\rho 2\pi\rho L \quad (1.95)$$

Hence,

$$\mathbf{E} = \frac{Q}{2\pi\varepsilon\rho L} \mathbf{a}_\rho \quad (1.96)$$

Neglecting flux fringing at the cylinder ends,

$$V = - \int_2^1 \mathbf{E} \cdot d\mathbf{l} = - \int_b^a \left[\frac{Q}{2\pi\epsilon\rho L} \mathbf{a}_\rho \right] \cdot d\rho \mathbf{a}_\rho \quad (1.97)$$

$$= \frac{Q}{2\pi\epsilon L} \ln \frac{b}{a} \quad (1.98)$$

Thus the capacitance of a coaxial cylinder is given by

$$C = \frac{Q}{V} = \frac{2\pi\epsilon L}{\ln \left(\frac{b}{a} \right)} \quad (1.99)$$

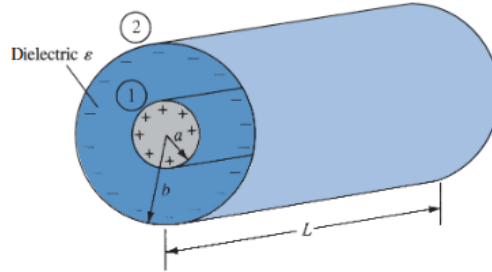


Figure 1.14: A coaxial capacitor.

C. Spherical Capacitor

A spherical capacitor is the case of two concentric spherical conductors. Consider the inner sphere of radius a and outer sphere of radius b ($b > a$) separated by a dielectric medium with permittivity ϵ as shown in Figure 1.15. We assume charges $+Q$ and $-Q$ on the inner and outer spheres, respectively. By applying Gauss's law to an arbitrary Gaussian spherical surface of radius r ($a < r < b$), we have

$$Q = \epsilon \oint_S \mathbf{E} \cdot d\mathbf{S} = \epsilon E_r 4\pi r^2 \quad (1.100)$$

that is,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \quad (1.101)$$

The potential difference between the conductors is

$$V = - \int_2^1 \mathbf{E} \cdot d\mathbf{l} = - \int_b^a \left[\frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \right] \cdot dr \mathbf{a}_r \quad (1.102)$$

$$= \frac{Q}{4\pi\epsilon} \left[\frac{1}{a} - \frac{1}{b} \right] \quad (1.103)$$

Thus the capacitance of the spherical capacitor is

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}} \quad (1.104)$$

By letting $b \rightarrow \infty$, $C = 4\pi\epsilon a$, which is the capacitance of a spherical capacitor whose outer plate is infinitely large. Such is the case of a spherical conductor at a large distance from other conducting bodies—the isolated sphere. Even an irregularly shaped object of about the same size as the sphere will have nearly the same capacitance. This fact is useful in estimating the stray capacitance of an isolated body or piece of equipment.

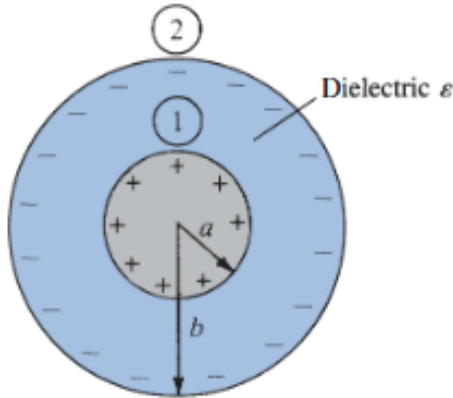


Figure 1.15: A spherical capacitor.

Example: Conducting Spherical Shells

Conducting spherical shells with radii $a = 10$ cm and $b = 30$ cm are maintained at a potential difference of 100 V such that $V(r = b) = 0$ and $V(r = a) = 100$ V. Determine V and $\mathbf{E}$ in the region between the shells. If $\epsilon_r = 2.5$ in the region, determine the total charge induced on the shells and the capacitance of the capacitor.

Solution Consider the spherical shells shown in Fig.1.16 and assume that V depends only on r . Hence Laplace's equation becomes

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0 \quad (1.105)$$

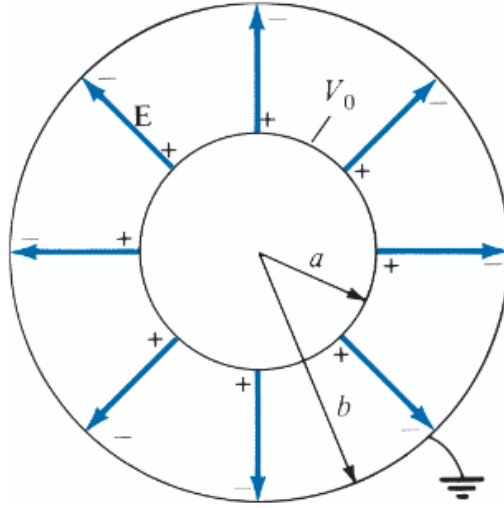


Figure 1.16: Potential $V(r)$ due to conducting spherical shells

Since $r \neq 0$ in the region of interest, multiply through by r^2 :

$$\frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0 \quad (1.106)$$

Integrating once gives

$$r^2 \frac{dV}{dr} = A \quad (1.107)$$

or

$$\frac{dV}{dr} = \frac{A}{r^2} \quad (1.108)$$

Integrating again gives

$$V = -\frac{A}{r} + B \quad (1.109)$$

As usual, constants A and B are determined from the boundary conditions.
When $r = b$, $V = 0$:

$$0 = -\frac{A}{b} + B \quad \Rightarrow \quad B = \frac{A}{b} \quad (1.110)$$

Hence,

$$V = A \left(\frac{1}{b} - \frac{1}{r} \right) \quad (1.111)$$

Also when $r = a$, $V = V_0$:

$$V_0 = A \left(\frac{1}{b} - \frac{1}{a} \right) \quad (1.112)$$

or

$$A = \frac{V_0}{\frac{1}{b} - \frac{1}{a}} \quad (1.113)$$

Thus,

$$V = V_0 \frac{\left(\frac{1}{r} - \frac{1}{b} \right)}{\left(\frac{1}{a} - \frac{1}{b} \right)} \quad (1.114)$$

The electric field is obtained from

$$\mathbf{E} = -\nabla V = -\frac{dV}{dr} \mathbf{a}_r = -\frac{A}{r^2} \mathbf{a}_r \quad (1.115)$$

Substituting the value of A from eq. (1.113),

$$\mathbf{E} = \frac{V_0}{r^2 \left(\frac{1}{a} - \frac{1}{b} \right)} \mathbf{a}_r \quad (1.116)$$

The total charge on the inner shell is

$$Q = \oint_S \varepsilon \mathbf{E} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^\pi \varepsilon \frac{V_0}{a^2 \left(\frac{1}{a} - \frac{1}{b} \right)} a^2 \sin \theta \, d\theta \, d\phi \quad (1.117)$$

$$Q = \frac{4\pi\varepsilon V_0}{\left(\frac{1}{a} - \frac{1}{b} \right)} \quad (1.118)$$

Alternatively,

$$\rho_s = D_n = \varepsilon E_n, \quad Q = \int_S \rho_s dS$$

The capacitance is easily determined as

$$C = \frac{Q}{V_0} = \frac{4\pi\varepsilon}{\left(\frac{1}{a} - \frac{1}{b}\right)} \quad (1.119)$$

which is the same result obtained earlier.

Substituting $a = 0.1$ m, $b = 0.3$ m, and $V_0 = 100$ V gives

$$V = 100 \left[\frac{\frac{1}{r} - \frac{10}{3}}{10 - \frac{10}{3}} \right] = 15 \left[\frac{1}{r} - \frac{10}{3} \right] \text{ V} \quad (1.120)$$

Check

$$\nabla^2 V = 0, \quad V(0.3 \text{ m}) = 0, \quad V(0.1 \text{ m}) = 100 \text{ V}$$

$$\mathbf{E} = \frac{100}{r^2(10 - \frac{10}{3})} \mathbf{a}_r = \frac{15}{r^2} \mathbf{a}_r \text{ V/m} \quad (1.121)$$

$$Q = \pm 4\pi \left(\frac{10^{-9}}{36\pi} \right) (2.5)(100) \left(\frac{1}{10 - \frac{10}{3}} \right) = \pm 4.167 \text{ nC} \quad (1.122)$$

The positive charge is induced on the inner shell; the negative charge is induced on the outer shell.

$$C = \frac{|Q|}{V_0} = \frac{4.167 \times 10^{-9}}{100} = 41.67 \text{ pF} \quad (1.123)$$

1.9 Energy Density in Electrostatic Fields

To determine the energy present in an assembly of charges, we must first determine the amount of work necessary to assemble them. Suppose we wish to position three point charges Q_1 , Q_2 , and Q_3 in an initially empty space shown in Figure 1.17. No work is required to transfer Q_1 from infinity to P_1 because the space is initially charge free and there is no electric field [from earlier equation, $W = 0$]. The work done in transferring Q_2 from infinity to P_2 is equal to the product of Q_2 and the potential V_{21} at P_2 due to Q_1 . Similarly, the work done in positioning Q_3 at P_3 is equal to $Q_3(V_{32} + V_{31})$, where V_{32} and V_{31} are the potentials at P_3 due to Q_2 and Q_1 , respectively.

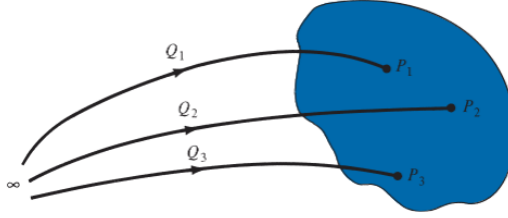


Figure 1.17: Assembling of charges

Hence the total work done in positioning the three charges is

$$W_E = W_1 + W_2 + W_3 = 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32}) \quad (1.124)$$

If the charges were positioned in reverse order,

$$W_E = W_3 + W_2 + W_1 = 0 + Q_2 V_{23} + Q_1 (V_{12} + V_{13}) \quad (1.125)$$

where V_{23} is the potential at P_2 due to Q_3 , V_{12} and V_{13} are, respectively, the potentials at P_1 due to Q_2 and Q_3 . Adding eqs. (1.124) and (1.125) gives

$$2W_E = Q_1 (V_{12} + V_{13}) + Q_2 (V_{21} + V_{23}) + Q_3 (V_{31} + V_{32}) \quad (1.126)$$

$$= Q_1 V_1 + Q_2 V_2 + Q_3 V_3 \quad (1.127)$$

or

$$W_E = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3) \quad (1.128)$$

where V_1 , V_2 , and V_3 are total potentials at P_1 , P_2 , and P_3 , respectively. In general, if there are n point charges, eq. (1.128) becomes

$$W_E = \frac{1}{2} \sum_{k=1}^n Q_k V_k \quad (\text{in joules}) \quad (1.129)$$

If, instead of point charges, the region has a continuous charge distribution, the summation in eq. (1.129) becomes integration; that is,

$$W_E = \frac{1}{2} \int_L \rho_l V dl \quad (\text{line charge}) \quad (1.130)$$

$$W_E = \frac{1}{2} \int_S \rho_s V dS \quad (\text{surface charge}) \quad (1.131)$$

$$W_E = \frac{1}{2} \int_V \rho_v V dv \quad (\text{volume charge}) \quad (1.132)$$

Since $\rho_v = \nabla \cdot \mathbf{D}$, eq. (1.132) can be further developed to yield

$$W_E = \frac{1}{2} \int_V (\nabla \cdot \mathbf{D}) V dv \quad (1.133)$$

But for any vector $\mathbf{A}$ and scalar V , the identity

$$\nabla \cdot (V\mathbf{A}) = \mathbf{A} \cdot \nabla V + V(\nabla \cdot \mathbf{A}) \quad (1.134)$$

or

$$(\nabla \cdot \mathbf{A})V = \nabla \cdot (V\mathbf{A}) - \mathbf{A} \cdot \nabla V \quad (1.135)$$

holds. Applying the identity in eq. (1.135) to (1.133), we get

$$W_E = \frac{1}{2} \int_V \nabla \cdot (V\mathbf{D}) dv - \frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv \quad (1.136)$$

By applying the divergence theorem to the first term on the right-hand side of this equation, we have

$$W_E = \frac{1}{2} \oint_S (V\mathbf{D}) \cdot d\mathbf{S} - \frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv \quad (1.137)$$

From last section, we recall that V varies as $1/r$ and $\mathbf{D}$ as $1/r^2$ for point charges; V varies as $1/r^2$ and $\mathbf{D}$ as $1/r^3$ for dipoles; and so on. Hence, $V\mathbf{D}$ in the first term on the right-hand side of eq. (1.137) must vary at least as $1/r^3$ while dS varies as r^2 . Consequently, the first integral in eq. (1.137) must tend to zero as the surface S becomes large. Hence, eq. (1.137) reduces to

$$W_E = -\frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv = \frac{1}{2} \int_V (\mathbf{D} \cdot \mathbf{E}) dv \quad (1.138)$$

and since $\mathbf{E} = -\nabla V$ and $\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E}$, the electrostatic energy is

$$W_E = \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \int_V \varepsilon_0 \varepsilon_r E^2 dv \quad (1.139)$$

From this, we can define electrostatic energy density w_E (in J/m³) as

$$w_E = \frac{dW_E}{dv} = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \varepsilon_0 \varepsilon_r E^2 = \frac{D^2}{2\varepsilon_0} \quad (1.140)$$

so eq. (1.137) may be written as

$$W_E = \int_V w_E dv \quad (1.141)$$

Boundary Conditions

So far, we have considered the existence of the electric field in a homogeneous medium. If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called *boundary conditions*. These conditions are helpful in determining the field on one side of the boundary if the field on the other side is known. Obviously, the conditions will be dictated by the types of material the media are made of. We shall consider the boundary conditions at an interface separating

- Dielectric (ε_{r1}) and dielectric (ε_{r2})
- Conductor and dielectric
- Conductor and free space

To determine the boundary conditions, we need to use Maxwell's equations:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0 \quad (1.142)$$

and

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} \quad (1.143)$$

where Q_{enc} is the free charge enclosed by the surface S . Also we need to decompose the electric field intensity $\mathbf{E}$ into two orthogonal components:

$$\mathbf{E} = \mathbf{E}_t + \mathbf{E}_n \quad (1.144)$$

where $\mathbf{E}_t$ and $\mathbf{E}_n$ are, respectively, the tangential and normal components of $\mathbf{E}$ to the interface of interest. A similar decomposition can be done for the electric flux density $\mathbf{D}$.

A. Dielectric–Dielectric Boundary Conditions

Consider the $\mathbf{E}$ field existing in a region that consists of two different dielectrics characterized by $\varepsilon_1 = \varepsilon_0 \varepsilon_{r1}$ and $\varepsilon_2 = \varepsilon_0 \varepsilon_{r2}$ as shown in Figure 1.18(a). The fields $\mathbf{E}_1$ and $\mathbf{E}_2$ in media 1 and 2, respectively, can be decomposed as

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n} \quad (1.145)$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n} \quad (1.146)$$

We apply eq. (1.142) to the closed path $abcd$ of Figure 1.18(a), assuming that the path is very small with respect to the spatial variation of $\mathbf{E}$. We obtain

$$0 = E_{1t}\Delta w - E_{1n}\frac{\Delta h}{2} - E_{2n}\frac{\Delta h}{2} - E_{2t}\Delta w + E_{2n}\frac{\Delta h}{2} + E_{1n}\frac{\Delta h}{2} \quad (1.147)$$

where $E_t = |\mathbf{E}_t|$ and $E_n = |\mathbf{E}_n|$. The $\frac{\Delta h}{2}$ terms cancel, and eq. (1.147) becomes

$$0 = (E_{1t} - E_{2t})\Delta w \quad (1.148)$$

or

$$E_{1t} = E_{2t} \quad (1.149)$$

Thus the tangential components of $\mathbf{E}$ are the same on the two sides of the boundary. In other words, $\mathbf{E}_t$ undergoes no change on the boundary and it is said to be continuous across the boundary. Since $\mathbf{D} = \varepsilon\mathbf{E} = \mathbf{D}_t + \mathbf{D}_n$, eq. (1.149) can be written as

$$\frac{D_{1t}}{\varepsilon_1} = E_{1t} = E_{2t} = \frac{D_{2t}}{\varepsilon_2} \quad (1.150)$$

or

$$\frac{D_{1t}}{\varepsilon_1} = \frac{D_{2t}}{\varepsilon_2} \quad (1.151)$$

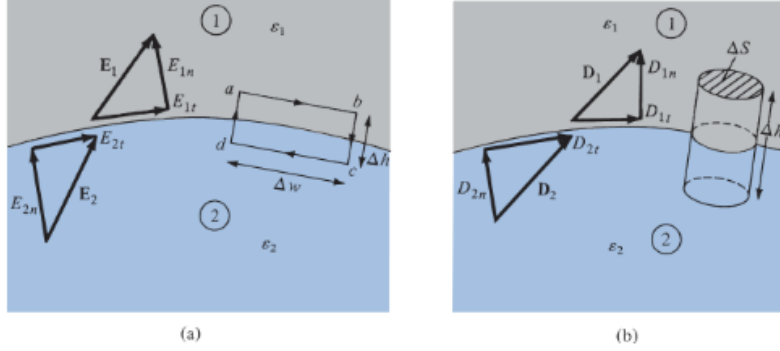


Figure 1.18: Dielectric-dielectric boundary: (a) determining $E_{1t} = E_{2t}$, (b) determining $D_{1n} = D_{2n}$

that is, D_t undergoes some change across the interface. Hence D_t is said to be discontinuous across the interface.

Similarly, we apply eq. (1.143) to the pillbox (cylindrical Gaussian surface) of Figure 1.18(b). The contribution due to the sides vanishes. Allowing $\Delta h \rightarrow 0$ gives

$$\Delta Q = \rho_s \Delta S = D_{1n} \Delta S - D_{2n} \Delta S \quad (1.152)$$

or

$$D_{1n} - D_{2n} = \rho_s \quad (1.153)$$

where ρ_s is the free charge density placed deliberately at the boundary. It should be borne in mind that eq. (1.153) is based on the assumption that $\mathbf{D}$ is directed from region 2 to region 1 and eq. (1.153) must be applied accordingly. If no free charges exist at the interface (i.e., charges are not deliberately placed there), $\rho_s = 0$ and eq. (1.153) becomes

$$D_{1n} = D_{2n} \quad (1.154)$$

Thus the normal component of $\mathbf{D}$ is continuous across the interface; that is, D_n undergoes no change at the boundary. Since $\mathbf{D} = \epsilon \mathbf{E}$, eq. (1.154) can be written as

$$\epsilon_1 E_{1n} = \epsilon_2 E_{2n} \quad (1.155)$$

showing that the normal component of $\mathbf{E}$ is discontinuous at the boundary. Equations (1.149) and (1.153) (or 1.154) are collectively referred to as *boundary conditions*; they must be satisfied by an electric field at the boundary separating two different dielectrics.

As mentioned earlier, the boundary conditions are usually applied in finding the electric field on one side of the boundary given the field on the other side. Besides this, we can use the boundary conditions to determine the *refraction* of the electric field across the interface. Consider $\mathbf{D}_1$ or $\mathbf{E}_1$ and $\mathbf{D}_2$ or $\mathbf{E}_2$ making angles θ_1 and θ_2 with the normal to the interface as illustrated in Figure 1.19. Using eq. (1.149), we have

$$E_1 \sin \theta_1 = E_{1t} = E_{2t} = E_2 \sin \theta_2 \quad (1.156)$$

$$E_1 \sin \theta_1 = E_2 \sin \theta_2 \quad (1.157)$$

Similarly, by applying eq. (1.154) or (1.155), we get

$$\varepsilon_1 E_1 \cos \theta_1 = D_{1n} = D_{2n} = \varepsilon_2 E_2 \cos \theta_2 \quad (1.158)$$

or

$$\varepsilon_1 E_1 \cos \theta_1 = \varepsilon_2 E_2 \cos \theta_2 \quad (1.159)$$

Dividing eq. (1.157) by eq. (1.159) gives

$$\frac{\tan \theta_1}{\varepsilon_1} = \frac{\tan \theta_2}{\varepsilon_2} \quad (1.160)$$

Since $\varepsilon_1 = \varepsilon_0 \varepsilon_{r1}$ and $\varepsilon_2 = \varepsilon_0 \varepsilon_{r2}$, eq. (1.160) becomes

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}} \quad (1.161)$$

This is the law of refraction of the electric field at a boundary free of charge (since $\rho_s = 0$ is assumed at the interface). Thus, in general, an interface between two dielectrics produces bending of the flux lines as a result of unequal polarization charges that accumulate on the opposite sides of the interface.

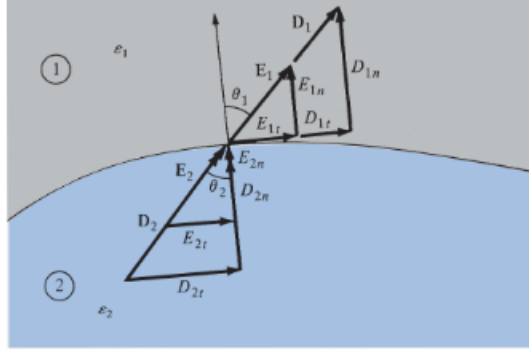


Figure 1.19: Refraction of $\mathbf{D}$ or $\mathbf{E}$ at a dielectric–dielectric boundary

B. Conductor–Dielectric Boundary Conditions

Figure 1.20 shows the case of conductor–dielectric boundary conditions. The conductor is assumed to be perfect (i.e., $\sigma \rightarrow \infty$ or $\rho_c \rightarrow 0$). Although such a conductor is not realizable for most practical purposes, we may regard conductors such as copper and silver as though they were perfect conductors.

To determine the boundary conditions for a conductor–dielectric interface, we follow the same procedure used for the dielectric–dielectric interface except that we incorporate the fact that $\mathbf{E} = 0$ inside the conductor. Applying eq. (1.142) to the closed path $abcd$ of Figure 1.20(a) gives

$$0 = 0 \cdot \Delta w + 0 \cdot \frac{\Delta h}{2} + E_n \frac{\Delta h}{2} - E_t \Delta w - E_n \frac{\Delta h}{2} - 0 \cdot \frac{\Delta h}{2} \quad (1.162)$$

As $\Delta h \rightarrow 0$,

$$E_t = 0 \quad (1.163)$$

Similarly, by applying eq. (1.143) to the cylindrical pillbox of Figure 1.20(b) and letting $\Delta h \rightarrow 0$, we get

$$\Delta Q = D_n \Delta S - 0 \cdot \Delta S \quad (1.164)$$

because $\mathbf{D} = \varepsilon \mathbf{E} = 0$ inside the conductor. Equation (1.164) may be written as

$$D_n = \frac{\Delta Q}{\Delta S} = \rho_s \quad (1.165)$$

or

$$D_n = \rho_s \quad (1.166)$$

Thus under static conditions, the following conclusions can be made about a perfect conductor:

1. No electric field may exist within a conductor; that is,

$$\rho_v = 0, \quad \mathbf{E} = 0 \quad (1.167)$$

2. Since $\mathbf{E} = -\nabla V = 0$, there can be no potential difference between any two points in the conductor; that is, a conductor is an equipotential body.
3. An external field $\mathbf{E}$ must be external to the conductor and must be normal to its surface; that is,

$$D_t = \varepsilon_0 \varepsilon_r E_t = 0, \quad D_n = \varepsilon_0 \varepsilon_r E_n = \rho_s \quad (1.168)$$

An important application of the fact that $\mathbf{E} = 0$ inside a conductor is in electrostatic screening or shielding. If conductor A kept at zero potential surrounds conductor B as shown in Figure 1.21, B is said to be electrically screened by A from other electric circuits, such as conductor C , outside A . Similarly, conductor C outside A is screened by A from B . Thus conductor A acts like a screen or shield, and the electrical conditions inside and outside the screen are completely independent of each other.

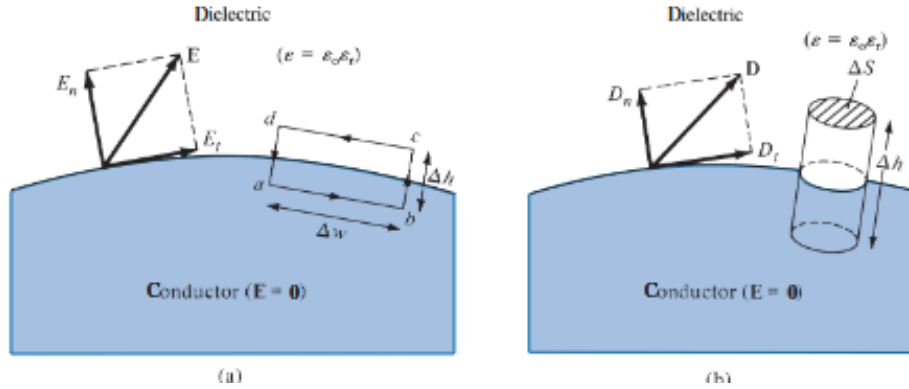


Figure 1.20: Conductor–dielectric boundary

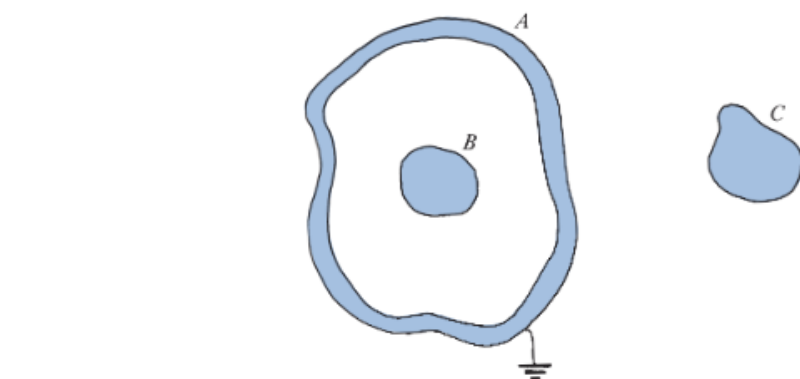


Figure 1.21: Electrostatic screening